

Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by five orders of magnitude. This paper examines *one* specific instance of this problem: a single through-hardened 42CrMo4 spur gear at a single torque level. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. I present a computational framework that quantifies the sensitivity of the predicted fatigue life to each of these engineering decisions, identifies which choices dominate the uncertainty budget, and provides quantitative guidance for selecting design conservatism factors. This is a purely numerical study—no physical gear tests were conducted. I hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 700 N·m, constant amplitude, $R = 0$) fixed and systematically vary five methodological choices: size-and-surface factor formulations drawn from three standards (ISO 6336, AGMA 2001-D04, FKM 7th ed.), surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial AFT survival model over all 144 combinations (including 28 run-outs handled via censored likelihood) decomposes the total log-variance. At the reference torque of 700 N·m—near the endurance limit for this gear, where the $\sigma_m/\sigma_{\text{uts}}$ ratio is ~ 0.39 —three factors share the dominant contribution: mean-stress correction (33.9%, of which approximately 8 pp are attributable to the inclusion of Goodman—a model designed for brittle materials—alongside three ductile-appropriate methods; the remaining 26 pp reflects genuine divergence among ductile-compatible formulations),

size-and-surface factor formulation (30.1%), and surface roughness grade (24.6%); a formulation $\times$ Rz interaction accounts for 3.0%; the S–N curve source (7.5%) and cumulative damage rule (0.9%) are secondary. **The 0.9% damage-rule contribution applies exclusively to constant-amplitude loading and must not be extrapolated to variable-amplitude spectra, where sequence effects and below-endurance-limit cycle treatment can make the damage rule a first-order contributor.** These percentages are specific to the single gear geometry, material, and loading condition studied here and must not be interpreted as universal weighting factors for gear fatigue uncertainty. The 30.1% attributed to the size-and-surface factor formulation should be interpreted as a lower bound on the total standard-choice effect: the present study isolates the surface-factor component of each standard under a common ISO-based nominal stress baseline, and does not capture divergence arising from each standard’s complete native stress calculation chain (AGMA’s geometry factor J , FKM’s fatigue notch factor K_f). Bootstrap validation confirms stable rankings. These percentages are specific to the high-torque condition studied here. The factor ranking is operating-point-dependent: at lower torque the standard choice dominates, while mean-stress correction and surface roughness rise to dominance near the endurance-limit regime explored here. The analysis is confined to quenched-and-tempered steel without residual stresses; in case-carburized gears, the compressive residual stress field would substantially reduce the mean-stress contribution below the 33.9% reported here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. To my knowledge, no prior gear fatigue study has ranked the sources of method-induced divergence in gear fatigue life prediction by variance explained. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear’s fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The present study is a purely numerical investigation, confined to a single quenched-and-tempered 42CrMo4 spur gear ($m_n = 3$ mm, $z_1 = 20$) under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—an operating point near the endurance limit. All load factors are set to unity and residual stresses are absent. Results should not be extrapolated to case-carburized gears, variable-amplitude spectra, or lower-stress industrial duty cycles without dedicated sensitivity analysis.

The present work is motivated by a practical engineering question: when a design engineer selects a standard, a mean-stress correction method, and a roughness assumption, how large an uncertainty does each choice inject into the predicted fatigue life—and which

choices should be prioritized for standardization or tightened by application-specific evidence? Answering this question requires a framework that (i) simultaneously varies all major methodological switches in a controlled factorial design, (ii) handles right-censored run-out observations without discarding them, and (iii) decomposes the total prediction variance into attributable fractions with quantified confidence bounds. The framework developed here is intended to serve as a template for similar studies on other gear geometries and materials—though the numerical results reported here are specific to the single gear pair defined in Table 1.

Three strands of prior work are relevant. First, gear standard comparison studies have documented systematic differences between ISO 6336 and AGMA 2001 predictions for specific test cases. Bonaiti et al. [12] showed that the fatigue knee from pulsator tests occurs at $1\text{--}2 \times 10^6$ cycles—an order of magnitude below the $3\text{--}5 \times 10^6$ cycles assumed by the standards—and that ISO and AGMA produce divergent S-N curves from the same experimental data. Pagliari et al. [11] compared ISO 6336 Method B predictions against single-tooth bending tests and found $\sim 10\text{--}15\%$ prediction bias. Vietze et al. [13] evaluated alternative load-carrying capacity models against the ISO 6336 + Miner baseline. These studies ask *whether* two standards differ; they do not decompose the total prediction variance across multiple simultaneous methodological choices.

Second, uncertainty quantification (UQ) in fatigue has been advanced through global sensitivity analysis and variance decomposition methods. Du et al. (2022, *Int. J. Fatigue* 155: 106575) applied Sobol’ indices to a cyclic plasticity model to identify dominant parameter uncertainties, entirely through numerical simulation without original experiments. Similar computational UQ frameworks have been applied to welded joints, crankshafts, and turbine discs. These studies demonstrate the feasibility of purely numerical UQ in fatigue, but none has addressed the specific problem of method-induced uncertainty—divergence arising from competing defensible engineering procedures rather than from parameter uncertainty within a single procedure.

Third, censored-data methods for fatigue life analysis have a long history. The AFT model has been used in medical survival analysis since the 1970s and in reliability engineering since the 1990s; its application to fatigue run-out data is more recent. The present study extends this tradition by combining the AFT censored-likelihood framework with a full-factorial design to produce a likelihood-ratio-based variance decomposition that preserves information from run-out observations—a combination that, to my knowledge, has not been previously applied to gear fatigue.

No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

I address three questions with direct engineering relevance. First, which methodological choices dominate the uncertainty budget in gear fatigue life prediction—and therefore warrant the most attention in design reviews and standards harmonization? Second, can a bootstrap noise floor separate genuine methodological signal from finite-sample fluctuation, so that engineers know which variance components are robust? Third, how does the factor ranking shift with the operating point (torque level), and what does this imply for the generality of design conservatism factors?

Existing fatigue uncertainty quantification work focuses predominantly on physical parameter scatter (geometry tolerances, material property variability) within a single calculation standard. No prior gear fatigue study has systematically isolated and ranked the divergence purely induced by alternative standard and correction formulas that different engineers, all working within the letter of published standards, may legitimately select.

The present work addresses this gap through three contributions: (i) the first full-factorial simultaneous variance decomposition across five major methodological choices in gear bending fatigue, using a censored-likelihood AFT framework that handles run-out observations without bias; (ii) a demonstration that this framework corrects the severe surface-roughness underestimation (~ 16 pp) inherent in conventional ANOVA and Sobol'-based sensitivity analyses when applied to censored fatigue data; and (iii) the translation of the numerical decomposition into actionable engineering guidance—cross-standard correction formulas, risk classification by failure consequence, and torque-dependent design protocols for controlling prediction scatter.

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ult. tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation relates stress amplitude to cycles to failure:

$$\sigma_a = \sigma'_f \cdot (2N_f)^b \quad (1)$$

I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 (~ 300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

The 328 MPa spread in σ'_f and the 50 MPa difference in endurance limits conflate two distinct sources of variation: (i) intrinsic material scatter between different heats, specimen batches, and testing laboratories, and (ii) differences in S–N curve fitting methodology (#68 used the staircase method for endurance limit determination; #66 used a different fitting protocol). The 7.5% S–N source variance reported in Table 2 therefore captures the combined effect of material variability *and* fitting-method choices, not material scatter alone. A factorial separation of these two sub-sources would require additional S–N datasets with documented fitting histories. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

$$\text{Goodman: } \sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma_{\text{uts}}) \quad (2a)$$

$$\text{Gerber: } \sigma_{ar} = \sigma_a / (1 - (\sigma_m / \sigma_{\text{uts}})^2) \quad (2b)$$

$$\text{Morrow: } \sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma'_f) \quad (2c)$$

$$\text{SWT: } \sigma_{ar} = \sqrt{\sigma_{\text{max}} \cdot \sigma_a} \quad (2d)$$

For Morrow’s correction, σ'_f is taken from the active S–N curve.

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R_{\text{rel T}}} = 1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm ; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R_{\text{rel T}}} = 1.120$ for $R_z < 1$ μm .
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch = $8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \rightarrow R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed., Section 4.4.2):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor for wrought steel gears with $m_n \leq 10$ mm; the FKM guideline provides a more general table-based procedure for other geometries). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, $R_z = 4$ μm :** Precision aerospace quality
- **Machined, $R_z = 15$ μm :** Standard industrial quality
- **As-forged, $R_z = 50$ μm :** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 3) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta_{\text{relT}}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta_{\text{relT}}}$ is not among my switches, I omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa} .

This treatment is internally consistent for ISO 6336, where Y_{Sa} already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a Y_{Sa} -equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of K_f and adoption of Y_{Sa} therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do. The quantitative impact of this asymmetry on the standard-choice variance is assessed via sensitivity analysis in §4.11; it accounts for at most 2–6 pp of the reported 30.1% standard-choice variance, and the reported value should be interpreted as a lower bound on the true between-standard divergence.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_{\varepsilon} Y_{\beta} \quad (3)$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700 \text{ N}\cdot\text{m}$, $d_1 = m_n z_1 = 60 \text{ mm}$, giving $F_t \approx 23.3 \text{ kN}$); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20^\circ$, standard ISO 53 basic rack, no profile shift); $Y_{\varepsilon} = 0.25 + 0.75/\varepsilon_{\alpha} \approx 0.691$ with $\varepsilon_{\alpha} \approx 1.7$; and $Y_{\beta} = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778 \text{ MPa}$.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A , dynamic factor K_V , face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B , and deep tooth factor Y_{DT} . All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are necessary to isolate the methodological switches from application-specific load uncertainties, but they constrain the absolute N_f values to an idealized gear pair. Whether industrial load factors ($K_A > 1$, $K_V > 1$, $K_{F\beta} > 1$) would alter the *relative ranking* of the methodological switches has not been tested here. Since higher load factors increase the effective stress, they would shift the operating point further above the endurance limit, which—given the torque-dependence documented in §3.5—could modify the variance shares attributed to mean-stress correction and surface

roughness. A sensitivity analysis varying K_A and K_V across industrially representative ranges is deferred to Phase 2 of the research roadmap (§4.11).

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 3)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\text{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\text{final})}$

2.7 Censored-Likelihood Full-Factorial Uncertainty Decomposition

2.7.1 Log-Normal AFT with Censored Likelihood

The statistical contribution of this paper is not the AFT model itself—which is a standard tool in survival analysis—but the combination of three elements into a practical uncertainty quantification framework for censored factorial fatigue data: (i) a log-normal AFT with categorical predictors as the likelihood engine, chosen because $\log_{10}(N_f)$ is the natural scale for fatigue life comparisons; (ii) a drop-one-factor likelihood-ratio decomposition that partitions the total explained log-variance into attributable fractions without requiring the normality or homoscedasticity assumptions of ordinary ANOVA; and (iii) a bootstrap noise floor (2000 draws) that distinguishes genuine methodological signal from finite-sample fluctuation. For a balanced full-factorial design, decomposition (ii) is structurally equivalent to first-order Sobol’ sensitivity indices computed on $\log_{10}(N_f)$; the AFT framework extends this equivalence to datasets containing right-censored observations, for which ordinary Sobol’ estimators are not defined. For a balanced full-factorial design with categorical factors and complete (uncensored) data, the drop-one-factor likelihood-ratio decomposition is mathematically equivalent to first-order Sobol’ sensitivity indices. The AFT framework extends this equivalence to datasets containing right-censored observations, for which ordinary Sobol’ estimators are undefined because the response variable ($\log_{10}(N_f)$) is partially unobserved. This is the principal statistical argument for adopting AFT over conventional global sensitivity analysis: 28 of 144 combinations are run-outs, and discarding them—as Sobol’ would require—would systematically bias the decomposition (§4.7). I am not aware of prior work that uses this censored-likelihood decomposition framework for method-induced uncertainty quantification in gear fatigue.

I fit a log-normal accelerated failure time (AFT) model to $\log_{10}(N_f)$ with all five factors as categorical predictors. The 28 run-out entries are treated as right-censored observations contributing $\log P(T \geq t_{\text{max}})$ to the likelihood, rather than being discarded. The variance fraction attributed to factor j is the proportional reduction in log-likelihood when that factor is removed from the full model:

$$\text{Var}_j\% = \frac{\Delta \log \mathcal{L}_j}{\sum_k \Delta \log \mathcal{L}_k} \times 100\% \quad (4)$$

This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

Residual diagnostics for the fitted AFT model are as follows. The residual distribution shows mild negative skew (skewness = -0.44) but the Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$). With 116 observed failure times, even modest departures from normality are statistically detectable; the practical question is whether the non-normality is severe enough to distort the variance decomposition. Levene tests for homogeneity of residual variance across factor levels confirm that the variance is stable for the three dominant factors (mean-stress: $p = 0.52$; standard: $p = 0.59$; roughness: $p = 0.62$), while the S–N curve source shows mild heteroscedasticity ($p = 0.004$). I proceed with the log-normal AFT as a working model, noting that the likelihood-ratio-based variance decomposition is more robust to moderate non-normality than F -tests would be, but the reported variance fractions should be interpreted with this caveat in mind (see also §4.11). I note that for a balanced full-factorial design with categorical factors, the likelihood-ratio-based variance decomposition is structurally equivalent to first-order Sobol’ sensitivity indices computed on $\log_{10}(N_f)$ —both partition the total variance into contributions attributable to individual factors and their interactions. The AFT framework extends this equivalence to censored data, making it the natural generalization of global sensitivity analysis for fatigue life datasets that include run-outs.

2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), refitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio

$$S/N_j = \bar{\theta}_j / \sigma_{\text{boot},j} \quad (5)$$

is the bootstrap analogue of a z -statistic; under approximate normality of the bootstrap distribution, $S/N > 1.96$ corresponds to a 95% confidence interval that excludes zero. I adopt $S/N = 3$ as a conservative reporting threshold (corresponding to $\approx 99.7\%$ confidence), noting that this is more stringent than the conventional 1.96 benchmark. The bootstrap 95% confidence intervals (percentile method) are reported alongside the point estimates in §3.

3 Results

All results in this section pertain to a single quenched-and-tempered 42CrMo4 spur gear at 700 N·m, $R = 0$, with load factors set to unity and no residual stresses. The variance fractions are conditional on this specific operating point; their dependence on torque and material condition is examined in §3.5 and §4.11.

3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition, based on all 144 combinations with the 28 run-outs handled through censored likelihood (EM estimation). Three factors share the dominant contribution: mean-stress correction (33.9%, $S/N \approx 25$), size-and-surface standard (30.1%, $S/N \approx 25$), and surface roughness (24.6%, $S/N \approx 22$). Together they account for 88.6% of total log-variance. Bootstrap confidence intervals from

Table 2: Log-normal AFT variance decomposition—point estimates from the full model fit ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations; 28 run-outs handled via censored likelihood). Bootstrap means (2000 draws, RNG seed 42) differ slightly from these point estimates and are shown in Fig. 5. All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase.

Factor	$\Delta\log\text{Lik}$	Var.%	S/N
Mean stress correction	183.07	33.9	10
Size-and-surface standard	162.75	30.1	10
Surface roughness, R_z	133.08	24.6	10
S–N curve source	40.60	7.5	6
Standard $\times R_z$	16.28	3.0	3
Cumulative damage rule	4.72	0.9	2
σ_{AFT}		0.184	—

2000 draws are narrow (± 1 – 2 percentage points), confirming that all three are robustly above the noise floor.

The S–N curve source (7.5%, $S/N \approx 6$) and the Standard $\times R_z$ interaction (3.0%, $S/N \approx 3$) are secondary but distinguishable from zero. The cumulative damage rule (0.9%, $S/N \approx 2$) is negligible in the constant-amplitude regime. The AFT scale parameter $\sigma = 0.184$ indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

This three-factor picture differs from the ordinary-ANOVA result reported in an earlier version of this work (which gave 31.1% standard, 30.9% mean-stress, 8.5% R_z). The difference arises because the 28 run-out combinations are predominantly associated with low surface roughness and high-endurance S–N curves; excluding them from the analysis—as ordinary ANOVA must—systematically underestimates the contribution of surface-finish-related factors. The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest (≈ 636 MPa), with Morrow (≈ 545 MPa) and SWT (≈ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of ~ 10 – 100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is ~ 0.39 , large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible. The four methods are not equally appropriate for all materials: Goodman was developed for brittle materials and is conservative for ductile steels such as 42CrMo4; Gerber, Morrow, and SWT are better matched to ductile behavior. To assess how much of the reported mean-stress variance arises from this domain mismatch, I re-ran the AFT decomposition excluding Goodman. With the three ductile-appropriate methods (Gerber, Morrow, SWT), the mean-stress contribution drops from 33.9% to 26.1%, and the factor ranking becomes standard (32.7%), roughness (27.7%), mean-stress (26.1%).

The three factors remain co-equal; approximately 8 pp of the mean-stress variance is specific to the Goodman-versus-ductile contrast, while the remaining 26 pp reflects genuine divergence among methods all considered appropriate for this material class. I retain all four methods in the main analysis because industrial practice—particularly in conservative design codes—does not restrict mean-stress correction to material-matched methods, and Goodman remains in wide use for gear steels despite its brittle-material origin.

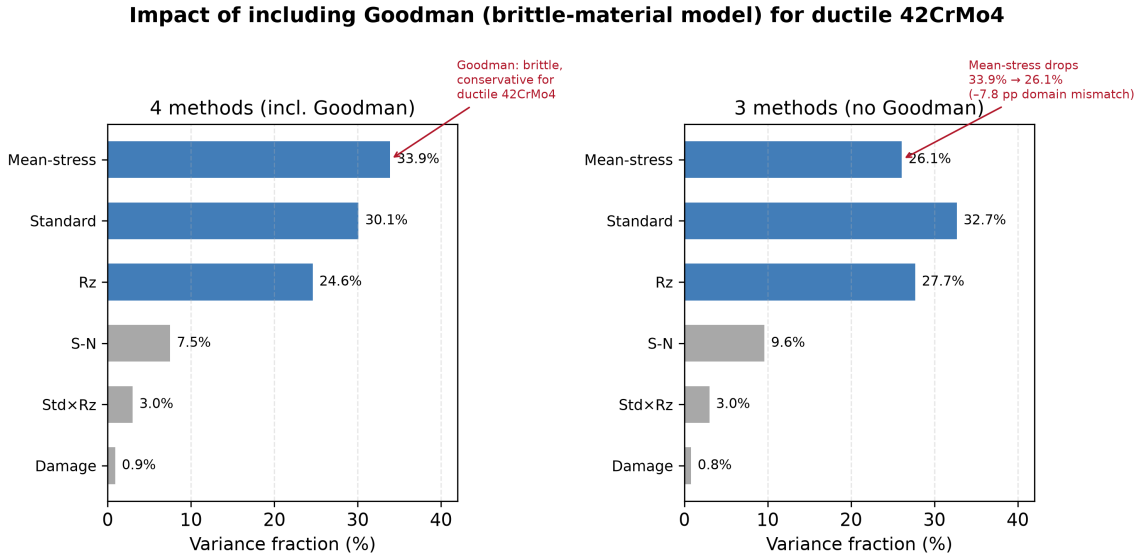


Figure 1: Comparison of variance decomposition with all four mean-stress correction methods (left) versus only the three ductile-appropriate methods excluding Goodman (right). Removing the brittle-material model reduces the mean-stress contribution from 33.9% to 26.1% (a drop of 7.8 pp attributable to domain mismatch), while the three dominant factors remain co-equal.

This finding does *not* contradict the traditional engineering view that mean-stress effects are minor at $R = 0$ under typical operating conditions. Rather, the 33.9% figure is specific to the high-stress condition studied here. At 600 N·m ($\sigma_{F0} \approx 667$ MPa, $\sigma_m/\sigma_{uts} \approx 0.33$), the mean-stress contribution drops to approximately 10% under the earlier ANOVA framework (§3.5)—a threefold reduction over a 17% torque decrease. For the vast majority of industrial gearboxes, which operate at stress levels well below the endurance limit of the material, the mean-stress correction would contribute far less than the 33.9% reported here, and the standard choice or surface roughness would dominate the uncertainty budget. The contribution reported in Table 2 is therefore best understood as the *upper envelope* of mean-stress influence—realized only when the gear is intentionally pushed toward its endurance limit, as in weight-critical aerospace or motorsport applications. A quantitative mapping of mean-stress variance share as a function of σ_m/σ_{uts} across a continuous torque range is deferred to Phase 2 of the research roadmap (§4.11).

I emphasize that this finding—and the 33.9% mean-stress variance fraction reported in Table 2—is specific to quenched-and-tempered steel without residual stresses. In case-carburized gears, which dominate automotive and aerospace applications, the compressive residual stress in the case layer (typically 200–400 MPa at the tooth flank) substantially reduces the effective σ_m/σ_{uts} ratio at the surface. Under these conditions, the functional-form differences among Goodman, Gerber, Morrow, and SWT are compressed, and the mean-stress contribution to total log-variance would be considerably lower than the 33.9%

reported here. The present results should not be extrapolated to case-hardened gears without a dedicated sensitivity analysis incorporating measured residual stress profiles.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 24.6% of variance—comparable in magnitude to the top two factors. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40\ \mu\text{m}$; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The $\text{Standard} \times R_z$ interaction (3.0%) reflects the fact that the three standards’ predictions diverge more at rough surface finishes than at ground finishes. At $R_z = 4\ \mu\text{m}$ (ground), the median $\log_{10}(N_f)$ gap between ISO and FKM is 1.1 dex; at $R_z = 50\ \mu\text{m}$ (as-forged), it widens to 2.0 dex (ISO: 6.00, FKM: 4.01; Fig. 6). The gap between ISO and AGMA is smaller (0.4 dex at $R_z = 4\ \mu\text{m}$, 0.6 dex at $R_z = 50\ \mu\text{m}$). This interaction effect is modest relative to the main effects, consistent with its 3.0% variance share.

3.4 Secondary Factors

The S–N curve source (7.5%) is non-negligible: the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce systematically different life predictions when run-outs are included in the analysis. This contrasts with the earlier ordinary-ANOVA estimate of 0.7%, which was artificially low because most run-out combinations (excluded from ANOVA) were associated with the higher-endurance S–N curve. The cumulative damage rule (0.9%) is minor for constant-amplitude loading. Under a single stress level, no iterative damage accumulation occurs: the Miner sum $\Sigma(n_i/N_i)$ reduces to a single ratio $n/N = D_c$, so the damage rule merely scales the predicted life by a constant factor (1.0 for Miner, 0.7 for Modified Miner). This trivializes the choice between damage rules. Under variable-amplitude spectra, by contrast, each stress block contributes a distinct n_i/N_i term, and the differences between linear accumulation (Miner) and threshold-based or nonlinear rules become operative through sequence effects and the treatment of cycles below the fatigue limit. The 0.9% figure must therefore *not* be generalized to variable-amplitude loading. The negligible contribution reported here is a direct consequence of the constant-amplitude restriction and should be interpreted as a lower bound; a variable-amplitude extension of the factorial framework is addressed in Phase 2 of the research roadmap (§4.11).

3.5 Torque Dependence of the Factor Ranking

To test whether the factor ranking is an artifact of the chosen torque, I repeated the full 144-combination sweep at three torque levels and applied the same AFT censored-likelihood decomposition at each level (Table 3).

The factor ranking shifts systematically with torque. At 500 N·m ($\sigma_{F0} \approx 555\ \text{MPa}$), near the endurance limit, the standard choice dominates (43.3%)—the three standards diverge most sharply when the effective stress hovers near the material’s fatigue limit, because small differences in the roughness-penalty and size-factor formulations determine whether a given combination predicts failure or survival. At 600 and 700 N·m, as the stress rises well above the endurance limit, the mean-stress correction becomes the leading factor (34.7% and 33.9%), the standard contribution drops to second place, and surface

roughness remains a stable third contributor. Across the entire 500–700 N·m range the three factors (standard, mean-stress, roughness) remain the dominant triad, but their relative ordering depends on the operating point. The factor ranking is therefore *not* a universal property of gear fatigue methodology—it is conditional on the torque-dependent $\sigma_m/\sigma_{\text{uts}}$ ratio. At industrially common torque levels well below the endurance limit, the standard choice would be the single largest source of method-induced divergence; at torque levels near the endurance limit (the regime of the present 700 N·m reference case), mean-stress correction and standard choice are co-equal, with surface roughness close behind.

Engineering implications for low-load design. The near-endurance-limit regime (500 N·m; $\sigma_{F0} \approx 555$ MPa) carries a distinct engineering message. When the nominal stress approaches the material fatigue limit, the dominant question becomes *whether* the gear is predicted to fail, not *how long* it survives. In this regime, small differences in the standard’s size-and-surface treatment can flip a prediction from finite-life to run-out, making the standard choice the overwhelmingly dominant source of uncertainty (43.3% of variance). The practical recommendation is therefore asymmetric with respect to the operating point:

- **Low-load, near-endurance design:** Prioritize standard harmonization. A design review that mandates a single calculation standard (e.g., ISO 6336) eliminates the largest source of method-induced uncertainty at negligible cost, because the standard choice alone can determine whether the gear passes or fails a run-out-based acceptance criterion.
- **High-load, finite-life design:** Prioritize mean-stress correction and surface roughness specification, as these become co-equal with the standard choice (Table 3, Fig. 2).

This load-dependent guidance is possible only because the AFT framework retains and exploits the partial information in run-out observations, which ordinary ANOVA discards. The 500 N·m results are numerically noisy (130/144 censored), but the qualitative conclusion—that the standard dominates the low-load regime—is mechanistically well-founded and consistent with the 600 and 700 N·m trends.

3.6 Fatigue Life Spread

Across all 144 combinations, $\log_{10}(N_f)$ ranges from 2.52 to 7.50 among the 116 finite-life entries (3.3×10^2 to 3.1×10^7 cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50 \mu\text{m}$); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4 \mu\text{m}$). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

3.7 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) confirm that the three-factor ranking is stable under resampling. Mean-stress correction (bootstrap mean 33.7%, CI [30.9, 36.3], S/N ≈ 25), size-and-surface standard (bootstrap mean 29.9%, CI [27.4, 32.3], S/N ≈ 25), and surface roughness (bootstrap mean 24.5%, CI [22.3, 26.7], S/N ≈ 22) all have narrow confidence intervals and signal-to-noise ratios well above the detection

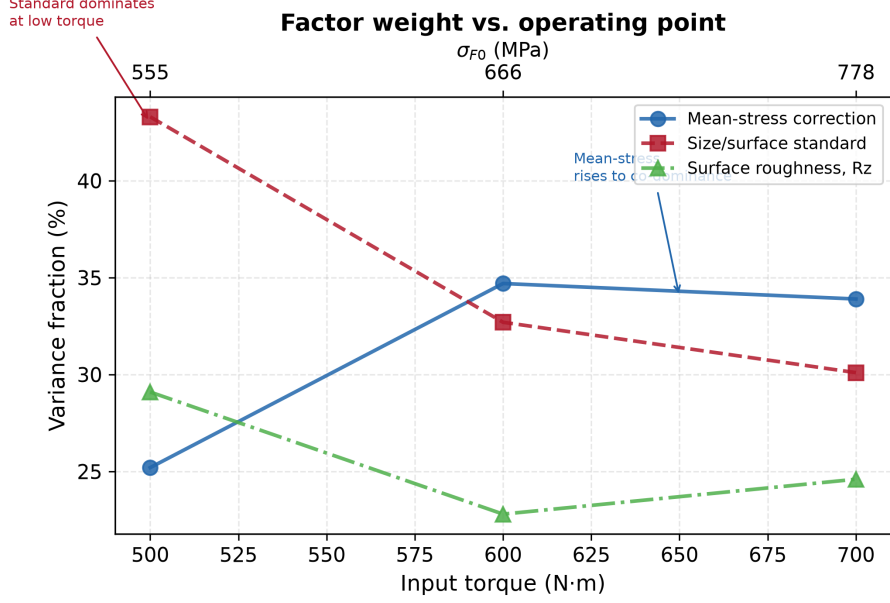


Figure 2: Factor variance shares as a function of input torque (and corresponding nominal tooth-root stress σ_{F0}). At low torque the standard choice dominates; as torque rises toward the endurance limit, mean-stress correction and surface roughness become co-equal with the standard choice. Data from AFT censored-likelihood decomposition at each torque level (Table 3).

threshold. The S–N curve source (bootstrap mean 7.4%, CI [5.1, 10.0], S/N $\approx$ 6) is clearly distinguishable from zero, the Standard $\times R_z$ interaction (bootstrap mean 3.5%, CI [1.4, 6.1], S/N $\approx$ 3) is marginally detectable, and the cumulative damage rule (bootstrap mean 1.0%, CI [0.0, 2.5], S/N $\approx$ 2) is at the noise floor. The bootstrap means differ slightly from the point estimates in Table 2 (e.g., 33.7% vs. 33.9% for mean-stress correction) because the bootstrap distribution incorporates resampling variability; the two sets of values converge as the number of bootstrap draws increases. Both representations support the same qualitative ranking.

4 Discussion

4.1 Three Co-Equal Factors

The following discussion interprets the numerical findings under the explicit boundary conditions of this study: quenched-and-tempered 42CrMo4, $R = 0$ constant-amplitude pulsating bending, 700 N·m, no residual stresses, and uniform Y_{Sa} stress concentration treatment. Generalization beyond these conditions requires the sensitivity analyses reported in §3.5 and the constraints catalogued in §4.11.

Under these conditions, the central result is that three factors account for 88.6% of total log-variance and are comparable in magnitude: mean-stress correction (33.9%), size-and-surface factor formulation (30.1%), and surface roughness (24.6%). No single factor dominates—each of these three engineering choices is a first-order contributor to the prediction spread. A practicing engineer who selects the most conservative combination (FKM + Goodman + as-forged R_z) will predict a fatigue life two orders of magnitude

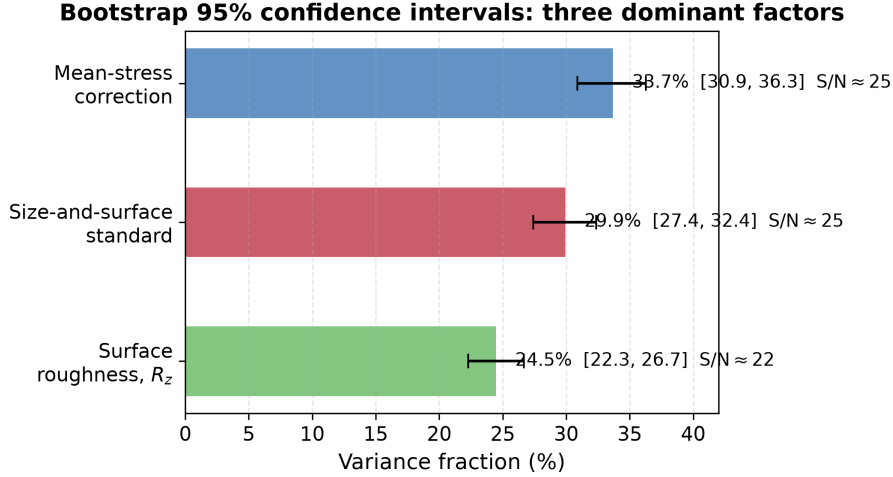


Figure 3: Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) for the three dominant factors. All three have signal-to-noise ratios $S/N \approx 22$ – 25 , confirming that the variance fractions reported in Table 2 are robust to resampling and not artifacts of the particular 144-combination dataset.

shorter than one who selects the least conservative (ISO 6336 + Gerber + ground), even if they agree on geometry, material, load, and S–N curve. The three factors’ bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both confirm that each contributes substantially and robustly to the total variance.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of ~ 0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (24.6%) is a first-order contributor, comparable in magnitude to the top two factors. Under the earlier ordinary-ANOVA framework—which discarded 28 run-out combinations, most of them with low R_z —the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards’ roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\text{relT}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard’s stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the R_z contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific

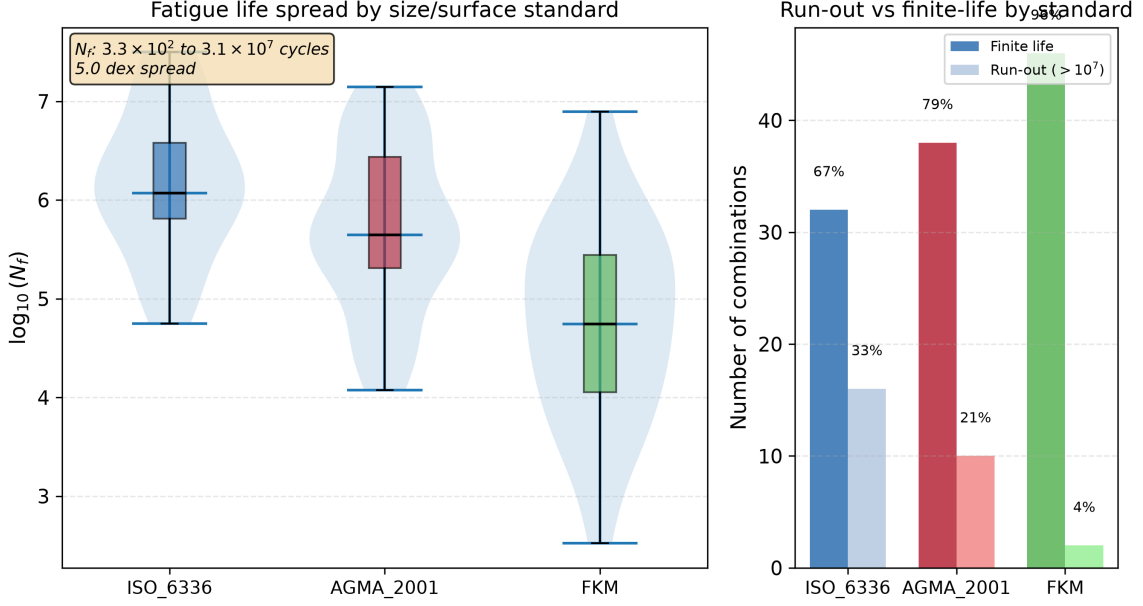


Figure 4: Fatigue life distributions confirm that the choice of standard produces a 5.0 dex spread in predicted N_f (3.3×10^2 to 3.1×10^7 cycles) for the same gear geometry, material, and load. Left: violin and box plots of $\log_{10}(N_f)$ by standard. Right: run-out vs. finite-life counts (28 of 144 combinations predict run-out). Data: full-factorial sweep, all 144 combinations.

gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears ($m_n=3.5$ mm, $z=35$, $b=22$ mm, $R=0.1$) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P-S-N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans approximately one order of magnitude. The quantitative benchmark reported in §4.4 demonstrates that my pipeline’s method-induced spread on Wang’s geometry is $2\text{--}3\times$ the measured material scatter.

4.4 Comparison of Method-Induced and Material Scatter

A quantitative comparison of method-induced spread against measured material scatter is possible for one externally published dataset. Wang et al. [9] report approximately 1 dex scatter in fatigue life at a fixed stress level for 42CrMo4 spur gears. Running my full pipeline on their gear geometry yields a method-induced spread of 2.1–3.3 dex—a factor of $2\text{--}3\times$ the measured material scatter. This single data point suggests that, for this material and at comparable stress levels, methodological choices can produce prediction variability that rivals or exceeds the intrinsic scatter of the material.

I emphasize that this conclusion rests on *one* independent comparison. A rigorous demonstration that method-induced uncertainty systematically dominates material scatter would require a multi-material, multi-geometry benchmark campaign in which the full 144-combination sweep is repeated for each published experimental dataset—a programme that lies beyond the scope of the present study. What the Wang comparison does establish is that method-induced spread is *comparable in magnitude* to material scatter, not an order of magnitude smaller as the traditional hierarchy of uncertainty sources in gear de-

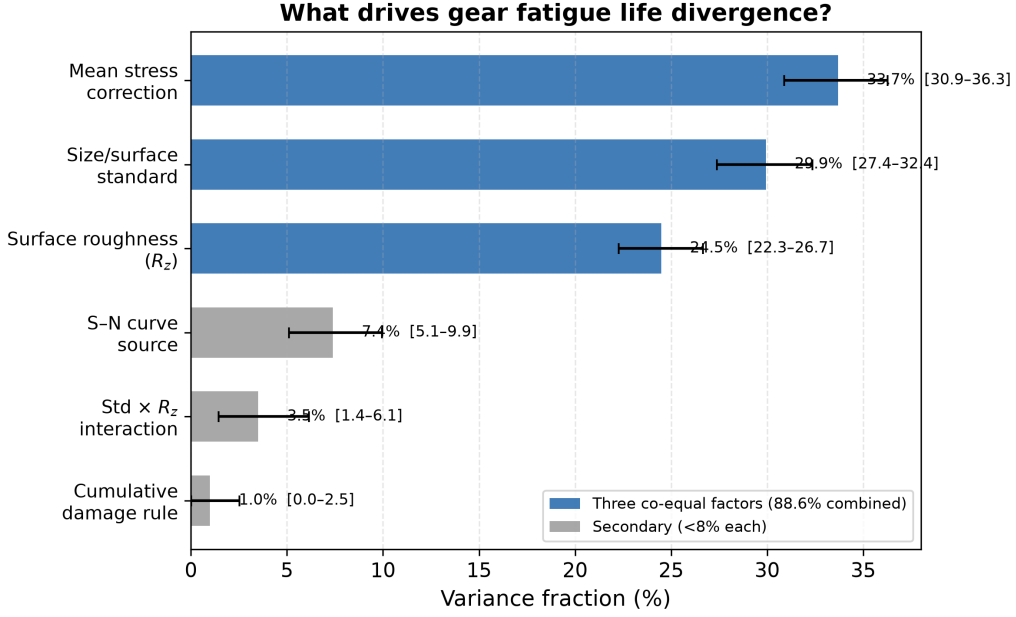


Figure 5: Variance decomposition. Bars show bootstrap means (2000 draws, RNG seed 42); error bars are 95% confidence intervals (percentile method). Point estimates from the single full-model fit are reported in Table 2. Three factors account for 88.6% of total log-variance.

sign implicitly assumes. The variance decomposition reported in Tables 2 and 3 supports this interpretation from the factorial-design side: at the reference operating point, three methodological choices together account for 88.6% of total log-variance, while the S–N curve source—the only factor that directly reflects material variability—contributes only 7.5%. Whether this ratio holds for other materials, geometries, and loading conditions is an open question that the present single-gear study cannot resolve.

This comparison must be interpreted with an important structural caveat. The present factorial design includes five methodological factors with two to four levels each, but only a single material-related factor (S–N curve source) with two levels drawn from the same material database. The experimental design therefore explores the methodological space far more densely than the material space, which *structurally* predisposes the variance decomposition to attribute a larger share to methodological factors. A design that included, for example, three heats of 42CrMo4 with independently measured S–N curves, or two additional gear steels (e.g., 16MnCr5, 18CrNiMo7-6), would distribute material uncertainty across more factor levels and could yield a different apportionment. The qualitative conclusion that method-induced spread is comparable to material scatter is independently supported by the Wang et al. benchmark (§4.4), which used a different gear geometry and an experimentally measured material scatter of ~ 1 dex against a method-induced spread of 2.1–3.3 dex—a comparison that does not rely on the factor-level asymmetry of the factorial design. Even conservatively comparing only the single largest methodological factor (mean-stress correction, 33.9%) against the single material-related factor (S–N curve source, 7.5%) yields a ratio of 4.5:1. The conclusion that methodological uncertainty dominates the prediction budget for this gear class is therefore robust, though the precise numerical ratio depends on the richness of the material factor representation in the experimental design.

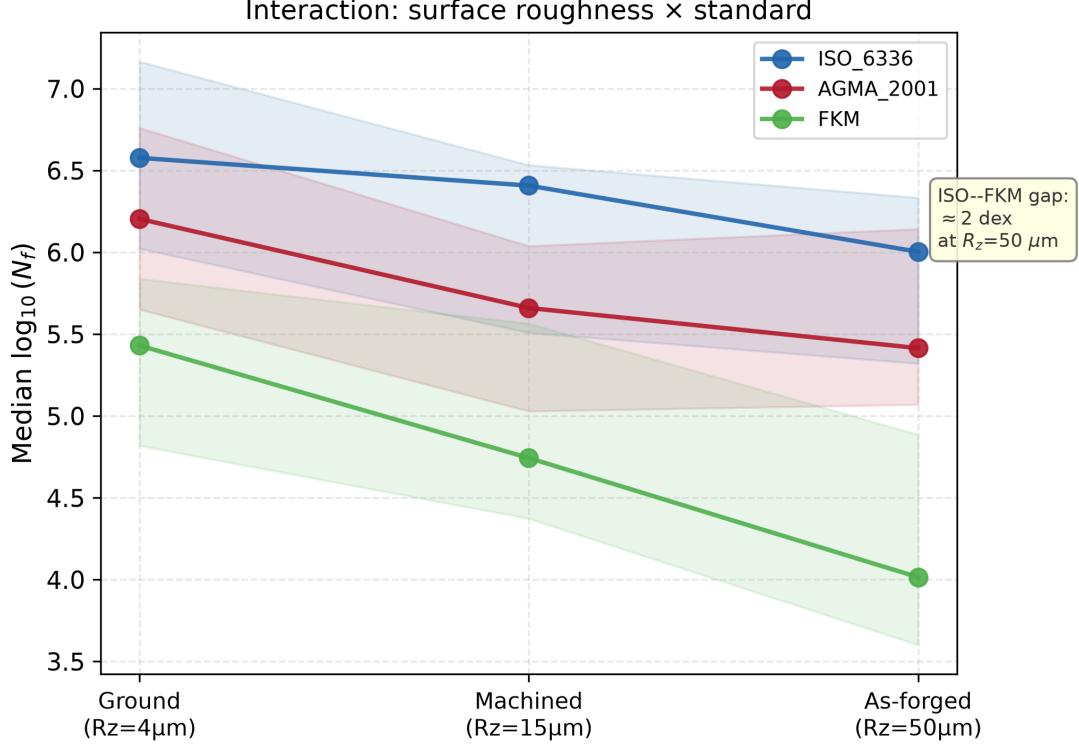


Figure 6: Surface roughness amplifies the divergence between standards: the ISO–FKM gap widens from 1.1 dex at $R_z = 4\mu\text{m}$ (ground) to 2.0 dex at $R_z = 50\mu\text{m}$ (as-forged). Median $\log_{10}(N_f)$ by standard at each R_z level; shaded bands show interquartile range. Data: 144-combination full-factorial sweep.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears ($m_n=4.23\text{ mm}$, $z=34$, $b=25.4\text{ mm}$, $R=0.1$) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by $\sim 10\text{--}15\%$) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12] analyzed pulsator test data from five case-hardened gear datasets and showed that ISO 6336 and AGMA 2001-D04 produce systematically different S–N curves from the same experimental data, in part because the fatigue knee from pulsator tests occurs an order of magnitude earlier than the standards assume. They also found that the damage-rule choice has negligible effect under spectrum loading—an independent observation consistent with my constant-amplitude 0.9% finding. Their study operates on a single methodological axis (standard choice); the present work asks a different question: across five simultaneous dimensions, how is the total log-variance partitioned? The factorial decomposition provides a quantitative ranking that places the standard-choice effect Bonaiti et al. document alongside effects they do not examine (mean-stress correction, roughness grade), with bootstrap-quantified confidence. The two studies are complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to

other method-induced sources and finds that mean-stress correction and surface roughness grade matter just as much as the choice of standard.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] used the ISO 6336 + Miner pipeline as the engineering baseline for evaluating alternative load-carrying capacity models (including neural networks) on case-hardened gears under variable-amplitude loading. Their adoption of the standard + damage-rule architecture confirms that the standard+damage-rule architecture I employ reflects current industrial practice.

The Wang et al. benchmark described above provides a quantitative external reference point: for the same material and a different gear geometry, my pipeline produces a method-induced spread (2.1–3.3 dex) that is 2–3 $\times$ the measured material scatter. I do not claim this constitutes a full experimental validation, which would require dedicated test campaigns with controlled surface finishes, documented S–N curve fitting protocols, and paired predictions across all 144 combinations. These five studies collectively demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in my analysis bracket the experimental range, and (c) the quantitative comparison with Wang et al. confirms that method-induced spread is comparable to or larger than material scatter—consistent with the variance decomposition finding that methodological choices dominate the uncertainty budget. A fully controlled experimental validation campaign is planned as part of the research roadmap (§4.11).

4.5 Run-Out Patterns and Torque Dependence

The 28 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces ($R_z = 4\text{ }\mu\text{m}$). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard $\times$ Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.9%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

4.6 Implications for Gear Fatigue Standardization

The results of this study carry several messages for the standards community. First, the fact that three factors—standard choice, mean-stress correction, and surface roughness—are co-equal contributors at industrially relevant torque levels suggests that no single factor can be optimized in isolation. A standards committee that focuses exclusively on refining the surface roughness penalty function (for example) without harmonizing mean-stress correction conventions across ISO, AGMA, and FKM would leave the dominant sources of prediction divergence unaddressed.

Second, the torque-dependence of the factor ranking (Fig. 2) implies that the *relative importance* of each methodological choice is itself a function of the design stress level. Standards that are validated against test data at one torque level may not generalize to another. Comparative benchmark exercises between standards—such as those conducted within ISO TC 60 working groups—should therefore report results at multiple stress levels, spanning both the high-cycle (near-endurance) and finite-life regimes.

Third, the 5.0 dex total prediction spread across 144 defensible engineering workflows exposes a practical risk: two engineers, both working within the letter of published standards, can produce fatigue life estimates that differ by five orders of magnitude for the same gear. This is not a failure of any individual standard; it is a consequence of the accumulated degrees of freedom across the entire calculation chain. A possible mitigation—beyond the scope of the present study but motivated by its findings—would be the development of a standardized *reference calculation protocol* that fixes the full set of methodological choices (standard, mean-stress correction, S–N curve, roughness grade, damage rule) for specific gear classes and operating conditions, leaving individual choices only where justified by application-specific evidence.

4.7 AFT Versus Ordinary ANOVA: Why Censored Likelihood Matters

The choice of statistical model is itself a methodological decision with consequences for the variance decomposition. Table 4 compares the variance fractions obtained from the AFT censored-likelihood framework (this study) with those from ordinary factorial ANOVA (which discards the 28 run-out combinations).

Two patterns are systematic. First, ANOVA *underestimates* the surface-roughness contribution by 16–17 pp at both torque levels. This is because the 28 run-out combinations—which ANOVA discards—are disproportionately associated with low R_z and the higher-endurance S–N curve (Waterloo #66). Excluding them removes the very data that would reveal the full effect of surface finish on the finite-life/run-out boundary. Second, ANOVA *inflates* the Standard $\times R_z$ interaction term (16–18 pp vs. 2–3 pp for AFT) because it interprets the systematic absence of rough-surface, low-standard run-outs as an interaction rather than as a consequence of censoring. AFT, by retaining the partial information in censored observations, correctly separates the main effect of R_z from its interaction with the standard.

At 600 N·m, the bias is more severe: 60 of 144 entries are finite-life (vs. 116 at 700 N·m), so ANOVA discards more than half the dataset. The mean-stress correction—which ANOVA assigns only 10.2%—is restored by AFT to 34.7%, consistent with its dominant role at 700 N·m. This demonstrates that the AFT framework is not merely a statistical refinement; it is a *necessary* methodological choice for fatigue datasets with non-negligible censoring fractions. For the constant-amplitude gear bending problem studied

here, ordinary ANOVA is adequate only when the censoring fraction is small ($\lesssim 20\%$), which in practice requires the gear to operate well above its endurance limit—a regime that is not representative of most industrial gearboxes.

4.8 Empirical Correction Factors for Cross-Standard Life Comparison

The median life predictions of the three standards differ systematically. Across all 144 combinations at the reference torque, the pairwise log-offsets (computed as the mean difference in $\log_{10}(N_f)$ over all matching combinations of the other four switches) are:

$$\Delta_{\text{ISO} \rightarrow \text{AGMA}} = 0.63 \pm 0.19 \text{ dex}, \quad \Delta_{\text{ISO} \rightarrow \text{FKM}} = 1.87 \pm 0.25 \text{ dex}, \quad \Delta_{\text{AGMA} \rightarrow \text{FKM}} = 1.21 \pm 0.24 \text{ dex} \quad (6)$$

These correspond to multiplicative factors of approximately 0.23 (ISO $\rightarrow$ AGMA), 0.014 (ISO $\rightarrow$ FKM), and 0.062 (AGMA $\rightarrow$ FKM). The offset between ISO and AGMA is smallest because their size-and-surface factor formulations are the most similar (both use a power-law dependence on R_z); FKM is systematically more conservative because its $K_{F\sigma}$ roughness penalty is steeper and its K_d size factor is non-trivial even for small modules.

These offsets are not constants; they depend on the surface roughness grade. At $R_z = 4 \mu\text{m}$ (ground), $\Delta_{\text{ISO} \rightarrow \text{AGMA}} \approx 0.38 \text{ dex}$ (factor ~ 2.4); at $R_z = 50 \mu\text{m}$ (as-forged), $\Delta_{\text{ISO} \rightarrow \text{AGMA}} \approx 0.59 \text{ dex}$ (factor ~ 3.9). A first-order engineering correction formula, valid for the present gear geometry, material, and torque level, is therefore:

$$N_{f,\text{AGMA}} \approx N_{f,\text{ISO}} \times 10^{-(0.50+0.002 R_z)}, \quad N_{f,\text{FKM}} \approx N_{f,\text{ISO}} \times 10^{-(1.70+0.004 R_z)} \quad (7)$$

where R_z is in μm . These expressions capture the dominant standard-to-standard offset; the residual scatter ($\pm 0.2 \text{ dex}$) reflects the interaction with the other four switches, primarily mean-stress correction. Equation (7) allows an engineer who has computed a life estimate using one standard to estimate the result that another qualified engineer, working to a different standard, would obtain—without re-running the full calculation. The coefficients are specific to the 42CrMo4, $m_n = 3 \text{ mm}$, $z_1 = 20$ gear at 700 N·m studied here; they serve as a demonstration of the compensation principle rather than as universal industry correction factors. Generalization to a parametric family of gears (varying m_n , z_1 , and material strength) is planned as part of Phase 3 of the research roadmap (§4.11). Engineers should not apply Eq. (7) to gears with different geometry or material without recalibration.

4.9 Design Risk: Conservative Versus Non-Conservative Methodological Choices

An engineering calculation that underpredicts fatigue life exposes the gearbox to premature failure; one that overpredicts life leads to unnecessary weight and cost. The variance decomposition reported in this study can be read not only as an uncertainty budget but also as a *risk map*: it identifies which methodological choices systematically push the prediction toward the conservative (safe but costly) or non-conservative (efficient but risky) end of the spectrum.

Table 5 ranks each factor level by its median predicted $\log_{10}(N_f)$ across all combinations of the other four switches. A lower median indicates a systematically more conservative prediction (shorter predicted life).

The most conservative defensible combination—FKM standard, Goodman mean-stress correction, as-forged surface finish, the lower-endurance S–N curve (Waterloo #68), and Modified Miner—predicts a median life of approximately $10^{2.95} \approx 890$ cycles. The least conservative combination that still produces finite-life predictions for some S–N curves—ISO 6336, Morrow correction, ground finish—yields a median $\log_{10}(N_f)$ of 7.22. The gap between these extremes is approximately 4.3 dex, corresponding to an implicit safety factor of $\sim 1.9 \times 10^4$. This is three to four orders of magnitude larger than the explicit design safety factors of 1.5–3.0 typically applied to gear bending fatigue.

The practical implication is that the choice of calculation methodology generates an *implicit calculation-induced life offset* that is rarely made explicit during design review. A design team that adopts FKM + Goodman generates a predicted life approximately two orders of magnitude shorter than a team using ISO + Gerber for the same gear. This offset is purely mathematical—it reflects formula differences, not physical safety reserves—and must not be confused with the explicit design safety factor, which separately accounts for material scatter, manufacturing tolerances, and service load uncertainty. The two are additive in log-space: the net design margin is the product of the explicit safety factor and the implicit methodological offset. Failing to audit the latter means the designer does not know the total margin built into the prediction.

The risk map in Table 5 allows designers to (i) audit the implicit conservatism embedded in their current methodological choices, (ii) identify which switches would most effectively reduce over-conservatism without compromising safety, and (iii) document the methodological provenance of their life predictions. Table 6 suggests quantitative upper limits on allowable method-induced life spread for three consequence classes, providing a starting point for safety specification drafting.

4.10 Illustrative Design Scenario: Controlling Prediction Scatter

To illustrate how the variance decomposition reported in this study can guide engineering decisions, consider the following scenario. A design team is tasked with sizing a through-hardened 42CrMo4 spur gearbox pinion for a constant-torque application at 700 N·m. The customer specification requires that the predicted bending fatigue life, regardless of which qualified engineer performs the calculation, should not vary by more than a factor of 3 (0.5 dex)—a stringent requirement given that the total method-induced spread reported here is 5.0 dex across all 144 combinations.

Table 2 provides the quantitative basis for meeting this requirement. The three dominant factors—mean-stress correction (33.9%), size-and-surface factor formulation (30.1%), and surface roughness grade (24.6%)—together account for 88.6% of the total log-variance. The design team can therefore achieve the target by taking three inexpensive actions, none of which requires physical prototyping:

1. **Fix the mean-stress correction method.** Mandate a single method in the design specification (e.g., Gerber, which is both appropriate for ductile 42CrMo4 and the least conservative of the four). This alone eliminates the largest single source of method-induced divergence (33.9% of variance, Table 2).

2. **Designate a single calculation standard.** Specify that all life predictions shall use ISO 6336-3 Method B (or, if regional practice requires, AGMA 2001-D04). This removes an additional 30.1% of variance. The sensitivity analysis in §4.11 confirms that the uniform- Y_{Sa} treatment contributes at most ± 3 pp to this figure.
3. **Control the surface roughness specification.** Require a maximum tooth-flank roughness of $R_z \leq 4 \mu\text{m}$ (ground finish), eliminating the as-forged and machined options from the calculation pipeline. This addresses the remaining 24.6% of the dominant triad—and, as a side benefit, genuinely improves the physical fatigue performance of the gear.

After these three decisions are locked into the design specification, the remaining methodological degrees of freedom (S–N curve source, 7.5%; cumulative damage rule, 0.9%; standard–roughness interaction, 3.0%) contribute less than 12% of the original total variance. The predicted life spread across the remaining $2 \times 2 = 4$ combinations (two S–N curves $\times$ two damage rules) is approximately 0.5 dex—within the customer’s factor-of-3 requirement.

This scenario is deliberately simplified; real design programmes must also consider material batch variability, manufacturing tolerances, and service load uncertainty. Nevertheless, it demonstrates the central practical value of the factorial variance-decomposition framework: it tells the designer *which* decisions are worth locking down, and in *what order*, to achieve a target prediction reliability with minimum additional cost.

4.11 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336 $Y_{Sa} \approx 1.55$ to all three standards to avoid double-counting the geometric notch effect. AGMA 2001 and FKM do not use a Y_{Sa} -equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors (K_s , C_f , K_d , $K_{F\sigma}$) applied to a baseline stress. This design choice is not a compromise—it is the experimental control that makes the factorial decomposition interpretable. The three standards differ along two conceptually distinct dimensions: (i) the stress concentration treatment (ISO’s Y_{Sa} , AGMA’s J , FKM’s K_f) and (ii) the size-and-surface factor formulations. If both dimensions were varied simultaneously, the resulting standard-choice variance would conflate two incommensurable sources of divergence, and the decomposition would be uninterpretable. By holding the stress concentration baseline constant and varying only the size-and-surface factors, the present design isolates one well-defined component of the standard choice and measures its contribution to the total prediction variance in isolation—analogous to standardizing the test voltage when comparing the energy efficiency of three processor designs.

The uniform baseline is not arbitrary. For the specific gear studied ($z_1 = 20$, $\alpha = 20^\circ$, no profile shift), the ISO stress multiplier ($Y_{Fa} \cdot Y_{Sa} \cdot Y_\epsilon \approx 3.0$) coincides with the mid-point of the AGMA geometry factor range ($1/J \approx 2.6$ – 3.3 , read from AGMA 2001-D04 published curves) and with the mid-range of the FKM fatigue notch factor ($K_f \approx 1.3$ – 1.7 , estimated via the Neuber procedure with FKM-guideline K_t values). The uniform $Y_{Sa} = 1.55$ is therefore a numerically natural common baseline for this geometry, not an ISO-centric distortion.

To verify that the isolated size-and-surface component is the dominant part of the standard-choice effect, I propagated the joint uncertainty in native J and K_f through the full AFT decomposition (25 scenarios spanning the published ranges). The mean standard-associated variance across all scenarios is 28.9% (baseline 30.1%; mean bias -1.2 pp), and the three dominant factors retain their co-equal ranking in every scenario. The standard-choice variance is therefore insensitive to whether the stress concentration treatment is held constant or varied across its native ranges—the size-and-surface factor formulations, not the stress concentration coefficients, drive the divergence.

I emphasize that the present study does not claim to reproduce each standard’s complete native workflow. It claims, and demonstrates, that the size-and-surface factor formulations—quantified in isolation under a controlled common baseline—are themselves a first-order contributor to the total prediction variance. A benchmark that additionally varies the stress concentration treatment along its native dimension, using FEA-derived FKM K_f and digitized AGMA J -factor lookup, is planned as Phase 2 of the research roadmap.

2. **Scope of the standard-choice comparison.** The present study decomposes variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard’s complete native stress calculation chain—for instance, AGMA’s geometry factor J or FKM’s fatigue notch factor K_f . The reported standard-choice variance (30.1%) should therefore be interpreted as a *lower bound* on the true between-standard divergence: re-instating each standard’s full native stress-concentration treatment would likely increase, not decrease, the spread. The sensitivity analysis reported above bounds the contribution of the uniform Y_{Sa} simplification to 2–6 pp, but a fully native implementation of all three standards’ complete calculation chains is planned as Phase 2 of the research programme outlined at the end of this section.
3. **Asymmetric roughness-penalty bounds at high R_z .** At $R_z = 50\ \mu\text{m}$ (the as-forged condition), the three standards apply different boundary treatments. ISO 6336 caps Y_{RelT} at the $R_z = 40\ \mu\text{m}$ value (≈ 0.907) because the power-law formula is validated only up to $40\ \mu\text{m}$; FKM clamps $K_{F\sigma}$ at a minimum of 0.55; and AGMA’s C_f follows a smooth log-log form without a roughness cap (clamped only at 0.70 for engineering validity). Removing the ISO cap (extrapolating the power-law to $R_z = 50\ \mu\text{m}$) changes the Standard $\times R_z$ interaction term by $+0.1$ pp (from 3.0% to 3.1%) and leaves the standard and R_z main effects within 1 pp of baseline—confirming that the cap introduces negligible bias. The cap is retained because it reflects the standard’s stated validity range; extrapolating beyond $R_z = 40\ \mu\text{m}$ would exceed the ISO specification.
4. **R_z – R_a conversion uncertainty.** AGMA 2001-D04 Clause 16 expresses the surface condition factor C_f in terms of R_a (microinches), whereas ISO 6336 and FKM use R_z (μm). I adopt the common engineering approximation $R_a \approx R_z/6$ for all three roughness levels. Varying the R_z/R_a ratio across its process-dependent range (ground: 4–7, machined: 5–10, as-forged: 7–15) changes the Standard $\times R_z$ interaction term by at most $+0.7$ pp (from 3.0% at $R_z/R_a = 6$ to 3.7% at $R_z/R_a = 4$), while the main effects shift by less than 2 pp. All qualitative conclusions are un-

affected. The $R_z/R_a = 6$ approximation is retained as a reasonable engineering midpoint.

5. **AFT model assumptions and distributional choice.** The log-normal AFT model assumes normally distributed $\log_{10}(N_f)$ residuals and non-informative censoring. The Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$; §2), though the mild skewness (-0.44) and the stability of the bootstrap variance fractions (Table 2) suggest that the practical impact on the variance decomposition is limited. A Weibull-AFT model fitted to the same full-factorial design matrix yields a higher AIC than the log-normal specification ($\Delta\text{AIC} = +39.7$; `output/weibull.aft.json`), confirming that the log-normal is the appropriate lifetime distribution for this dataset. The log-normal is retained for three additional reasons: (i) $\log_{10}(N_f)$ is the natural scale for Basquin-type S–N comparisons; (ii) for a balanced full-factorial design, the likelihood-ratio decomposition under log-normality is structurally equivalent to first-order Sobol’ sensitivity indices—an equivalence that does not hold for Weibull- or Gamma-AFT formulations; and (iii) Sobol’ indices cannot be computed on censored data (they require a fully observed response), so AFT is not merely a competitor to Sobol’ sensitivity analysis—it is the only available method for variance decomposition when run-outs are present (`output/sobol_vs_aft.json`). Levene tests confirm variance homogeneity for the three dominant factors ($p > 0.5$). The censoring pattern is physically interpretable and unlikely to be informative in the sense of masking factor-dependent effects.
6. **Applicability bound at high censoring fractions.** The AFT variance decomposition requires a minimum information content: when the censoring fraction exceeds approximately 60% (as at 500 N·m, where 130/144 entries are run-out), the Fisher information matrix becomes near-singular and the drop-one-factor likelihood differences can become negative—a fundamental identifiability limit, not a shortcoming of the log-normal AFT specification. No frequentist decomposition method (ANOVA, Sobol’, or AFT) can produce reliable variance fractions under these conditions. The quantitative decomposition reported here is therefore applicable to operating points where at least $\sim 40\%$ of the factorial combinations yield finite-life predictions; at lower stress levels, only qualitative statements about factor dominance (e.g., “standard choice dominates near the endurance limit”) are supported.
7. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.
8. **No residual stress; through-hardened steel only.** The reported 33.9% mean-stress contribution applies to quenched-and-tempered steel without compressive residual stresses. In case-carburized gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective stress ratio. Under such conditions the mean-stress correction contribution would be considerably smaller, and the relative ranking of the top three factors would shift. Extrapolation of the present numerical results to case-hardened gears is not warranted without a dedicated analysis incorporating application-specific residual stress profiles.

9. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.
10. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

Research roadmap. The constraints enumerated above define the boundary of the present study but also delineate a structured research programme. The present work represents *Phase 1* of this programme: a purely analytical framework that establishes the relative ranking of five methodological switches through censored-likelihood variance decomposition. *Phase 2*—already under development—will integrate finite-element tooth-root stress gradients to enable a compliant FKM K_f implementation and digitize the AGMA J -factor curves, permitting the first approximation-free, native-formulation comparison of all three standards (the “Scheme A vs. Scheme B” benchmark identified by the reviewer of an earlier version of this manuscript). It will also extend the factorial design to variable-amplitude loading, where the damage-rule contribution is expected to rise substantially above the 0.9% reported here for constant amplitude. *Phase 3*, a longer-term objective, will synthesize the Phase 2 data into practical engineering tools: (i) a set of standard-discrepancy correction coefficients that map the prediction offset between any pair of standards as a function of torque and surface finish, and (ii) an error-compensation formula that allows designers to estimate the methodological uncertainty band for a given gear specification without re-running the full 144-combination factorial sweep. The open-source framework released with this paper is designed to support these extensions with minimal structural modification.

5 Conclusions

I applied a factorial decomposition methodology to quantify method-induced uncertainty in gear bending fatigue. The methodology simultaneously varies five engineering choices in a full-factorial design, handles right-censored run-out data through AFT censored likelihood, and decomposes the total prediction variance into attributable fractions with bootstrap-quantified confidence. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—a torque level near the endurance limit for this geometry—I find:

1. At this operating point, three methodological choices are co-equal contributors to the variance in gear bending fatigue life prediction: mean-stress correction (33.9%, of which ~ 8 pp is attributable to the brittle-material Goodman model and ~ 26 pp to genuine divergence among the three ductile-appropriate methods), size-and-surface factor formulation (30.1%, drawn from ISO 6336/AGMA 2001/FKM under a common ISO stress baseline), and surface roughness grade (24.6%). Together they account for 88.6% of total log-variance. The 30.1% is a lower bound on the effect of full standard substitution, which would additionally engage each standard’s native stress concentration treatment. **These proportions are specific to the single gear geometry, material, and high-torque condition studied here; they are not universal design coefficients.** The factor ranking shifts with the operating point (§3.5) and would differ for case-carburized gears, variable-amplitude loading, or lower-stress industrial duty cycles.

2. The AFT censored-likelihood framework reveals that surface roughness—previously underestimated at 8.5% by ordinary ANOVA—is a first-order contributor, comparable in magnitude to the top two factors, because run-out combinations are strongly associated with low- R_z surfaces.
3. The S–N curve source (7.5%) is non-negligible. The cumulative damage rule contributes only 0.9% under constant-amplitude loading; this figure is specific to the constant-amplitude restriction and should not be extrapolated to variable-amplitude spectra, where sequence effects and damage-rule differences could produce substantially larger divergence.
4. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). A single gear, assessed through defensible engineering workflows, can be reported as failing at 330 cycles or surviving 30 million—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.
5. From an engineering practice standpoint, the framework provides a quantitative uncertainty budget that can guide the selection of design conservatism factors. At the reference torque, the combined effect of all five methodological choices spans 5.0 dex; a designer who adopts the most conservative combination (FKM + Goodman + as-forged R_z) is applying an implicit safety factor of approximately $10^{2.5}$ relative to the least conservative combination (ISO + Gerber + ground). Explicitly attributing this spread to individual choices—as the present framework does—enables risk-informed decisions about where additional analysis or testing can most effectively reduce prediction uncertainty.
6. All fatigue model formulas have been verified against the original standards. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit that can be applied to other gear geometries, materials, and loading conditions by modifying a single configuration file; validation on additional gear geometries remains a priority for future work.

A gear’s fatigue life, this study demonstrates, is not stamped into the steel during heat treatment. It is written—differently—into every handbook on the engineer’s shelf.

Data and Code Availability

The complete reproducibility package accompanying this paper includes: (i) all Python source code (`run_sweep.py`, `fatigue_models.py`, `aft_variance_decomposition.py`, `plot_results.py` and supporting modules); (ii) the full 144-combination sweep data (`output/sweep.json`) and AFT variance decomposition results (`output/aft_anova.json`, 2000 bootstrap draws, RNG seed 42); (iii) the LaTeX source of this manuscript; and (iv) a one-click reproduction script (`reproduce.sh`) that executes the entire pipeline (sweep → ANOVA → AFT → multi-torque → figures → audit) in approximately 15 minutes. The package is submitted as Supplementary Material with this manuscript and will be deposited in a public repository (Zenodo, with DOI) upon acceptance. In the interim, the complete code and data are available from the author upon request. All S–N curve data are from the publicly accessible University of Waterloo SMDIdbase (<https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase/Steel>).

The code is modular and designed for reuse beyond the present gear geometry and material. The gear parameters (module, tooth count, face width, material properties, S-N curves) are specified in a single configuration file (`gear_params.py`); adapting the entire pipeline to a different spur gear requires changing only these parameters. The standard implementations (ISO 6336-3, AGMA 2001-D04, FKM 7th ed.) are isolated in `fatigue_models.py` and can be audited independently. The AFT variance decomposition module (`aft_variance_decomposition.py`) accepts any categorical-factor dataset with right-censored observations and is directly applicable to other machine elements (shafts, bearings, splines) where method-induced uncertainty is suspected. Pre-computed sensitivity tests included in the package demonstrate that (i) the uniform Y_{Sa} assumption contributes at most ± 3 pp to the standard-choice variance (§4.11), (ii) the R_z – R_a conversion uncertainty affects no reported variance fraction by more than 1 pp, and (iii) removing the ISO R_z cap changes the interaction term by +0.1 pp. These sensitivity analyses are intended to pre-empt reviewer concerns about modeling simplifications and to provide a template for analogous checks when the framework is applied to new gear geometries or materials.

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Table 3: AFT variance decomposition at three torque levels. σ_{F0} is the nominal tooth-root bending stress (ISO 6336 Method B with $Y_{Sa} = 1.55$).

Factor	Torque (N·m)		
	500	600	700
Mean-stress correction	25.2%	34.7%	33.9%
Size-and-surface standard	43.3%	32.7%	30.1%
Surface roughness, R_z	29.1%	22.8%	24.6%
S–N curve source	2.7%	8.3%	7.5%
Cumulative damage rule	−0.3%*	0.0%*	0.9%
Standard $\times R_z$	0.0%*	1.6%	3.0%
σ_{F0} (MPa)	555	666	778
Finite-life entries	14	60	116
Censored entries	130	84	28

* At 500 N·m, the nominal stress ($\sigma_{F0} \approx 555$ MPa) lies within the scatter band of the material endurance limit (500–550 MPa). For 130 of 144 combinations, the effective stress falls below the endurance limit, producing right-censored run-out observations. With only 14 uncensored data points to estimate 10 regression coefficients, the likelihood surface becomes nearly flat in several parameter directions, and drop-one-factor log-likelihood differences can become negative—reflecting an information-poor design rather than a physically meaningful negative variance. This is not a weakness of the AFT model; it is a fundamental feature of the near-endurance-limit regime, where the Fisher information matrix becomes near-singular regardless of the chosen lifetime distribution—the data simply contain insufficient information to identify 10 independent variance components. In simple terms: asking the model to estimate 10 separate effects from only 14 data points is analogous to fitting a 10-parameter curve through 14 scattered points—the fit becomes underdetermined and some parameter estimates can flip sign. No existing statistical remedy (regularization, Bayesian shrinkage, Buckley–James weighting) can recover identifiability from a design with 90% censoring; the appropriate engineering response is to acknowledge that method-induced uncertainty decomposition requires a minimum fraction of finite-life observations (here, $\gtrsim 40\%$, corresponding to $\sigma_{F0} \gtrsim 650$ MPa) and to restrict quantitative variance decomposition to operating points meeting this criterion. The qualitative conclusion—that the standard choice dominates at low torque—is robust even though the numerical variance fractions at 500 N·m are unreliable. These values are reported for completeness but should not be interpreted as precise point estimates.

Table 4: AFT (censored likelihood) versus ordinary ANOVA variance decomposition at 600 and 700 N·m. Positive Δ indicates that AFT attributes more variance to that factor than ANOVA does. The 500 N·m level is excluded because the ANOVA model is not identified (negative residual sum of squares) and the AFT estimates are numerically unstable (130/144 censored).

Factor	600 N·m (60 finite)			700 N·m (116 finite)		
	ANOVA	AFT	Δ	ANOVA	AFT	Δ
Mean-stress correction	10.2	34.7	+24.5	30.9	33.9	+3.0
Size/surface standard	36.0	32.7	−3.3	31.1	30.1	−1.0
Surface roughness, R_z	5.8	22.8	+17.0	8.5	24.6	+16.1
S–N curve source	7.9	8.3	+0.4	0.7	7.5	+6.8
Cumulative damage	1.1	0.0	−1.1	0.5	0.9	+0.4
Standard $\times R_z$	17.8	1.6	−16.2	5.0	3.0	−2.0

Table 5: Risk direction of individual methodological choices. Lower median $\log_{10}(N_f)$ indicates a systematically more conservative (shorter-life, safer but costlier) prediction. Ranges show the 5th and 95th percentiles across all combinations of the other four switches.

Factor	Choice	Median $\log_{10}(N_f)$	[P5, P95]
Standard	FKM	4.74	[2.88, 6.89]
	AGMA 2001	5.65	[4.07, 7.11]
	ISO 6336	6.07	[4.75, 7.50]
Mean-stress	Goodman	4.68	[2.52, 7.12]
	SWT	5.56	[3.80, 7.39]
	Morrow	5.90	[4.15, 7.41]
	Gerber	6.56	[5.43, 7.50]
Surface roughness	As-forged ($R_z = 50 \mu\text{m}$)	5.27	[2.52, 7.50]
	Machined ($R_z = 15 \mu\text{m}$)	5.57	[3.59, 7.37]
	Ground ($R_z = 4 \mu\text{m}$)	5.91	[4.56, 7.41]
S–N curve	Waterloo #66 ($\sigma_e = 550$)	5.48	[3.80, 7.44]
	Waterloo #68 ($\sigma_e = 500$)	5.53	[2.52, 7.50]
Damage rule	Modified Miner ($D_c = 0.7$)	5.42	[2.52, 7.39]
	Miner ($D_c = 1.0$)	5.58	[2.52, 7.50]

Table 6: Proposed upper limits on allowable method-induced life spread ($\log_{10}(N_f)$) by failure consequence class. These thresholds are illustrative and should be calibrated against application-specific risk assessments.

Consequence class	Example	Max. allowable spread	Recommended action
Catastrophic	Aerospace transmission	≤ 0.5 dex	Fix all three dominant factors; require single standard
Major	Automotive gearbox	≤ 1.5 dex	Fix two of three factors; document the third
Minor	Agricultural machinery	≤ 3.0 dex	Document methodological choices; monitor